

POWER BI Q & A

1. What is Power BI?

Power BI is a Microsoft business analytics tool used for data visualization, reporting, and interactive dashboards.

2. What are the major components of Power BI?

Power BI Desktop, Power BI Service, Power BI Mobile App, Power BI Gateway, Power BI Report Server.

3. Difference between Power BI Desktop and Power BI Service?

Power BI Desktop is for report creation and designing; Power BI Service is a cloud platform for sharing and collaboration.

4. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI for calculated columns, measures, and custom calculations.

5. What is a Dashboard in Power BI?

A single-page canvas (one-page view) showing key metrics via tiles pinned from reports.

6. What is a Report in Power BI?

A multi-page data visualization document created in Power BI Desktop with charts, tables, slicers, etc.

7. What are DAX functions?

Predefined formulas like SUM(), CALCULATE(), FILTER(), RELATED() used to perform calculations on data.

8. What is a Measure in Power BI?

A dynamic calculation (like total sales or avg profit) that changes based on filters and slicers.

9. What is a Calculated Column?

A column computed at the data model level and stored in the table, created using DAX.

10. Difference between Measure and Calculated Column?

Calculated Column = Static, stored.

Measure = Dynamic, calculated on the fly based on filters.

11. What is Power Query?

Data transformation and cleaning tool in Power BI Desktop using M language.

12. What is ETL in Power BI?

Extract, Transform, Load process used to import and clean data before modeling.

13. What is a Relationship in Power BI?

A link between tables based on keys (One-to-Many, Many-to-One, etc.).

14. Types of Joins in Power Query?

Inner Join, Left Outer, Right Outer, Full Outer, Anti Join.

15. What is Star Schema?

Data model with a central fact table connected to multiple dimension tables.

16. What is Snowflake Schema?

Dimension tables are normalized into additional related tables forming a structure like a snowflake.

17. What is a Slicer?

A visual filter allowing users to filter data on reports (e.g., by date, region).

18. What are Filters in Power BI?

Visual, Page, and Report-level filters applied to control displayed data.

19. What is a KPI Visual?

Key Performance Indicator visual that shows progress toward a goal.

20. What is a Power BI Gateway?

A bridge that connects on-premises data sources to Power BI Service for refresh.

21. Types of Gateways?

- Personal Gateway
- On-premises Standard Gateway

22. What is Publish in Power BI?

Uploading a report from Power BI Desktop to Power BI Service.

23. Can we schedule data refresh in Power BI?

Yes, using Power BI Service with Gateway.

24. What is Row-Level Security (RLS)?

Restricts data access for specific users using roles and filters.

25. What is Drill-through?

Feature allowing navigation to a detailed report page focusing on selected data.

26. What is Drill-down?

Hierarchy navigation within visuals (Year → Quarter → Month).

27. What are Custom Visuals?

User-created or downloaded visuals from AppSource like Sankey, Heatmap, etc.

28. What is DirectQuery Mode?

Data stays in source; Power BI queries live without importing data.

29. Difference: Import vs DirectQuery?

- Import: Data is stored in PBIX file.
- DirectQuery: Live connection without storing data.

30. What is Composite Model?

A model that supports both Import and DirectQuery in a single report.

31. What is a Tooltip?

A small hover box showing additional data for visuals.

32. What is Bookmarks?

Feature to capture report view state (filters, visuals) for navigation.

33. What is Power BI Q&A?

Natural Language Query feature (Ask a question like “Total Sales by Region”).

34. What is Drillthrough Filter?

Filter allowing movement to detailed pages based on context.

35. What are Parameters in Power BI?

Dynamic inputs used in queries to control data extraction.

36. What is Performance Analyzer?

Tool to measure the performance of each visual and optimize.

37. What is the use of SUMX in DAX?

SUMX is an iterator function used to sum expressions row by row.

38. What is ALL() in DAX?

Removes all filter context from a table or column.

39. What is RELATED() in DAX?

Fetches related column values from another table using relationships.

40. What is CALCULATE() in DAX?

Modifies filter context and performs calculations in that context.

41. What is Power BI Service?

Cloud-based platform to share, collaborate, and manage Power BI reports.

42. Uses of Power BI Mobile?

View dashboards and reports on phone or tablet.

43. What is Paginated Report?

Pixel-perfect reports optimized for printing/export, created using Power BI Report Builder.

44. What is Workspace in Power BI Service?

Shared environment to store and share reports/dashboards.

45. What is App in Power BI?

Packaged collection of dashboards, reports shared with users.

46. What is Drill-up?

Moving to higher hierarchy level (Month to Year).

47. What file type is used by Power BI Desktop?

.pbix

48. Can Power BI connect to SQL Server?

Yes, via DirectQuery or Import mode.

49. What is M Language?

Formula language used in Power Query Editor for data transformations.

50. Why use Power BI over Excel?

Better data visualization, automation, large data handling, interactive dashboards, cloud sharing.

SQL Q&A

1. What is SQL?

SQL (Structured Query Language) is used to store, retrieve, manage, and manipulate data in relational databases.

2. What are the different types of SQL commands?

- **DDL** – Data Definition Language (CREATE, ALTER, DROP)
- **DML** – Data Manipulation Language (INSERT, UPDATE, DELETE)
- **DCL** – Data Control Language (GRANT, REVOKE)
- **TCL** – Transaction Control Language (COMMIT, ROLLBACK)

3. What is a Primary Key?

A unique identifier for each record in a table. Does not allow NULL or duplicate values.

4. What is a Foreign Key?

A field in one table that references the Primary Key of another table to create a relationship.

5. What is a Unique Key?

Ensures all values in a column are unique, but unlike primary key, unique key can allow **one NULL**.

6. What is the difference between WHERE and HAVING?

- **WHERE** filters rows before grouping.
- **HAVING** filters after GROUP BY.

7. What is a JOIN?

A JOIN is used to combine rows from multiple tables based on related columns.

8. Types of Joins in SQL?

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- CROSS JOIN

9. What is an INNER JOIN?

Returns only the matching rows between two tables.

10. What is a LEFT JOIN?

Returns all rows from left table + matched rows from right table.

11. What is a RIGHT JOIN?

Returns all rows from right table + matched rows from left table.

12. What is a FULL OUTER JOIN?

Returns all rows from both tables, matched or unmatched.

13. What is CROSS JOIN?

Returns **Cartesian product** – every row of table A with every row of table B.

14. What is a Self Join?

Joining a table with itself using aliases.

15. What is a View?

A virtual table created using an SQL query. It does not store data physically.

16. What is a Stored Procedure?

A saved SQL code that can be reused and executed multiple times.

17. What is a Function in SQL?

Returns a single value and can be used in SELECT statements.

18. Difference: Stored Procedure vs Function?

Stored Procedure	Function
Can return multiple values	Returns only one value
Supports DML	Mostly avoids DML
Uses EXEC	Used in SELECT

19. What is a Trigger?

An automatic action when INSERT, UPDATE, or DELETE is performed on a table.

20. What is Normalization?

Process of organizing data to reduce redundancy and improve integrity.

21. Types of Normalization Forms?

1NF, 2NF, 3NF, BCNF.

22. What is Denormalization?

Process of combining normalized tables to improve read performance.

23. What is a Constraint?

Rules applied to table columns (NOT NULL, UNIQUE, CHECK, DEFAULT).

24. What is the difference between DELETE and TRUNCATE?

DELETE	TRUNCATE
Removes specific rows	Removes all rows
Slower	Faster
Can be rolled back	Cannot be rolled back (in most DBs)

25. What is DROP?

Removes table structure completely.

26. What is GROUP BY?

Used to group rows with similar values for aggregate functions (SUM, COUNT).

27. What is ORDER BY?

Sorts the result in **ASC** (default) or **DESC** order.

28. What is an Index?

Improves search speed by creating a quick lookup reference.

29. Types of Indexes?

Clustered and Non-Clustered Index.

30. What is a Clustered Index?

Physically reorders table data. One per table.

31. What is a Non-Clustered Index?

Creates a separate structure, does not affect physical order.

32. What is SQL Injection?

Security vulnerability where malicious SQL is inserted into a query.

33. How to prevent SQL Injection?

Using parameterized queries and input validation.

34. What is DISTINCT?

Used to remove duplicate values from a result.

35. What is BETWEEN operator?

Filters values within a range (inclusive).

36. What is LIKE operator?

Used for pattern matching:

% = multiple characters, _ = one character.

37. What is IN operator?

Checks multiple values in WHERE using one condition.

38. What is UNION?

Combines results of two queries, removes duplicates.

39. What is UNION ALL?

Combines results including duplicates.

40. What is a Subquery?

A query inside another query.

41. What is a Correlated Subquery?

Subquery that depends on outer query for each row.

42. What is ACID in Database?

Atomicity, Consistency, Isolation, Durability – transaction properties.

43. What is a Transaction?

A group of SQL operations treated as a single unit.

44. Commands for Transaction Control?

COMMIT, ROLLBACK, SAVEPOINT.

45. What is EXISTS?

Checks if a subquery returns one or more rows.

46. What is COALESCE?

Returns first non-NULL value in the list.

47. What is NULL in SQL?

Represents missing or unknown data.

48. What is NVL or ISNULL?

Replaces NULL with a specified default value.

49. What is CURRENT_DATE or GETDATE()?

Returns current system date and time.

50. Difference: CHAR vs VARCHAR?

CHAR	VARCHAR
Fixed length	Variable length
Faster	Memory Efficient

1. What is Tableau?

Tableau is a leading data visualization tool used to create interactive dashboards and business intelligence reports.

2. What are the different Tableau products?

- Tableau Desktop
- Tableau Public
- Tableau Server
- Tableau Online
- Tableau Reader
- Tableau Prep

3. What is Tableau Desktop used for?

Used for creating reports, dashboards, and data visualization.

4. What is Tableau Public?

A free platform to create and publish visualizations online.

5. What is Tableau Server?

Enterprise platform used to share, publish, and manage dashboards securely.

6. What is Tableau Online?

A hosted (cloud) version of Tableau Server.

7. What is Tableau Reader?

A free tool to view Tableau dashboards locally (cannot edit).

8. What is Tableau Prep?

Used for data cleaning, shaping, and preparation before visualization.

9. What is a Dashboard in Tableau?

A collection of multiple visualizations on a single screen to analyze multiple views at once.

10. What is a Story in Tableau?

A sequence of dashboards or sheets used to present a data narrative.

11. What are Sheets in Tableau?

Worksheets used to create charts, maps, tables, etc.

12. What are Dimensions and Measures?

- **Dimensions:** Qualitative fields (Names, Dates, Category)
- **Measures:** Quantitative fields (Sales, Profit)

13. What is a Discrete field in Tableau?

Displays individual values (blue), like categories.

14. What is a Continuous field in Tableau?

Displays range of values (green), like numbers or dates.

15. What are Filters in Tableau?

Used to restrict data displayed in a view.

16. Types of Filters in Tableau?

- Extract Filter
- Data Source Filter
- Context Filter
- Dimension Filter
- Measure Filter

17. What is Context Filter?

A primary filter that affects other filters; improves performance.

18. What is Data Blending?

Combines data from multiple sources using a common field (primary & secondary data source).

19. Difference between Join and Blend?

Join

Blend

Performed in same data source Combines different data sources

Happens before aggregation Happens after aggregation

20. What is a Parameter in Tableau?

A dynamic input to replace constant values in filters, calculations, or reference lines.

21. What is a Calculated Field?

A custom field created using formulas for advanced calculations.

22. What is LOD Expression?

Level of Detail expressions are used to compute values at different data levels (e.g., FIXED, INCLUDE, EXCLUDE).

23. Types of LOD Expressions?

- **FIXED**
- **INCLUDE**
- **EXCLUDE**

24. What is Show Me in Tableau?

A panel suggesting visual types based on selected fields.

25. What is an Extract in Tableau?

A snapshot of data imported from the source for fast performance.

26. What is Live Connection?

Connects data in real-time from the source.

27. Difference: Extract vs Live?

Extract

Offline, high performance

Faster

Live

Real-time updates

Slower but always updated

28. What are Marks in Tableau?

Cards to control visual attributes (Color, Size, Label, Detail, Tooltip).

29. What is a Heat Map?

Chart showing data density using color intensity.

30. What is a Tree Map?

Displays hierarchical data as nested rectangles.

31. What is Dual Axis?

Allows combining two measures on a single graph with two axes.

32. What is a Bullet Chart?

Chart to compare actual vs target performance.

33. What is a Gantt Chart?

Used to show project timelines or duration.

34. What is a Histogram?

Represents distribution of continuous data.

35. What is a Trend Line?

A statistical line showing progression or prediction on data.

36. What is a Forecast in Tableau?

Predicts future values using built-in algorithm.

37. What is a Group in Tableau?

Combines similar members into a single category.

38. What is Set in Tableau?

Custom fields that define a subset of data.

39. What are Bins in Tableau?

Used to convert measures into ranges (e.g., age group 0–10, 10–20).

40. What is Data Densification?

Fills in missing data points in continuous graphs.

41. What is Page Shelf?

Used to analyze data one page at a time (animation-style).

42. Can Tableau connect to SQL databases?

Yes, Tableau supports SQL Server, MySQL, Oracle, PostgreSQL, etc.

43. What is a JOIN in Tableau?

Combining data tables using common fields (like SQL joins).

44. Types of JOINS in Tableau?

INNER, LEFT, RIGHT, FULL OUTER.

45. What is Tableau Public File Extension?

.twbx or .twb

46. What is Aggregation in Tableau?

Summarizing data like SUM, AVG, MIN, MAX.

47. What is Publish in Tableau?

Uploading dashboard to Tableau Server or Public to share.

48. What is Data Interpreter?

Helps clean Excel data by detecting headers & footers.

49. What is Dashboard Action?

Interactive feature for navigation (Filter, Highlight, URL Actions).

50. What are Performance Optimization Techniques?

- Use Extracts over Live
- Reduce Filters
- Use Context Filters
- Minimize complex calculations

1. What is Microsoft Excel?

Microsoft Excel is a spreadsheet software used for data entry, calculation, analysis, and visualization.

2. What are Excel formulas?

Formulas are expressions that perform calculations using values in cells. Example: =A1 + B1.

3. What are Excel functions?

Predefined formulas like SUM(), AVERAGE(), VLOOKUP(), etc.

4. What is the difference between Formula and Function?

Formula	Function
User-created expression	Predefined in Excel
Example: =A1+B1	Example: =SUM(A1:B1)

5. What is a Cell Reference?

The address of a cell (like A1, B2) used in formulas.

6. What are Absolute and Relative References?

- **Relative:** Changes when dragged (A1)
- **Absolute:** Fixed using \$ (\$A\$1)

7. What is VLOOKUP?

Used to search a value in the first column of a table and return a value from another column.

Example: =VLOOKUP(lookup_value, table_array, col_index, FALSE)

8. What is HLOOKUP?

Similar to VLOOKUP but searches horizontally across the first row.

9. What is Pivot Table?

A tool to summarize large data by grouping, filtering, and calculating totals/averages.

10. What is Conditional Formatting?

Applies formatting like color, bold, based on cell values (e.g., highlight values >100).

11. What is Data Validation?

Restricts data input in a cell (e.g., only numbers between 1-100).

12. What is IF function in Excel?

Checks a condition and returns a value.

Example: =IF(A1>50, "Pass", "Fail")

13. What are Charts/Graphs in Excel?

Visual representations of data like Bar, Line, Pie, Column charts.

14. What is CONCATENATE or CONCAT?

Joins multiple text strings.

Example: =CONCAT(A1, " ", B1)

15. What is SUMIF?

Adds cells based on a condition.

Example: =SUMIF(A:A, ">100", B:B)

16. What is COUNTIF?

Counts cells based on a single condition.

Example: =COUNTIF(A:A, "Yes")

17. What is the difference between COUNT and COUNTA?

COUNT

COUNTA

Counts numbers only Counts non-empty cells

18. What is Freeze Panes?

Locks rows/columns while scrolling.

19. What is a Macro in Excel?

Automates repetitive tasks using VBA (Visual Basic for Applications).

20. How do you remove duplicates in Excel?

Go to **Data → Remove Duplicates** to delete repeated values in a column or table.